

MICROSOFT POWERED HACKATHON

Microsoft Innovate 2026

*Innovate · Build · Empower***Team Name***XLR8***Team ID***491***Problem Statement Title***10,000 Reviews, No Time to Read Them***Theme / Bucket***Smart Assistants & Chatbots (Enterprise Conversational AI)***Team Leader***Sakham Kumar
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Mentor Evaluation

Team Details

Round 2 – Mentor Evaluation

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Problem Statement



What is the problem?

- Product teams receive more feedback than they can manually review each week.
- Important issues like bugs, UX problems, and pricing concerns often get overlooked.
- A team of 5 PMs can review *~200 responses/week*, while popular apps receive *10,000+ reviews* in the same period.



Who faces it?

- Product managers, customer support leads, and QA teams at any company with a public app or feedback channel.
- Occurs continuously — every release cycle, every day new reviews arrive on app stores, surveys, and support tickets.



Why do current options fall short?

- **Today:** manual spot-checking, basic keyword search, or generic star-rating dashboards.
- **Gap:** none of these surface why users are unhappy — they show sentiment scores but not traceable root causes.
- The problem persists because most tools summarize sentiment without linking back to real verbatims, so teams can't trust or act on the output.

CURRENT OPTIONS	LIMITATION
Keyword Search	Misses context
Star Ratings	No root cause
Sentiment Analysis	Doesn't link to evidence
Manual Review	Not scalable

Proposed Solution

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Our idea in one line

An AI-powered Feedback & Review Analyzer that transforms thousands of user reviews into actionable insights, helping product teams identify recurring issues, understand sentiment, and prioritize improvements faster.



How it solves the problem

- **Collects** reviews from app stores, surveys, and support channels automatically
- **Groups** similar feedback into meaningful themes such as bugs, pricing, and UX issues
- **Analyzes** sentiment to identify the most critical customer concerns quickly
- **Links** every insight to real user comments for easy verification and action



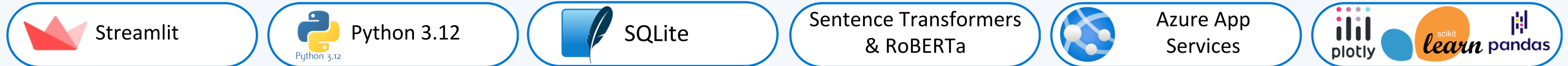
What makes it different

- **Enterprise-grade traceability:** Each detected theme is linked to its supporting user reviews, not just a sentiment score
- **Validated accuracy:** Sentiment predictions are checked against a labelled sample before the output is trusted
- **Privacy & reliability:** Built-in PII redaction and drift monitoring enable safe, continuous operation
- **Cost-effective implementation:** Uses free AI models and sentiment libraries (e.g., Hugging Face Transformers) with Power BI or a lightweight dashboarding tool

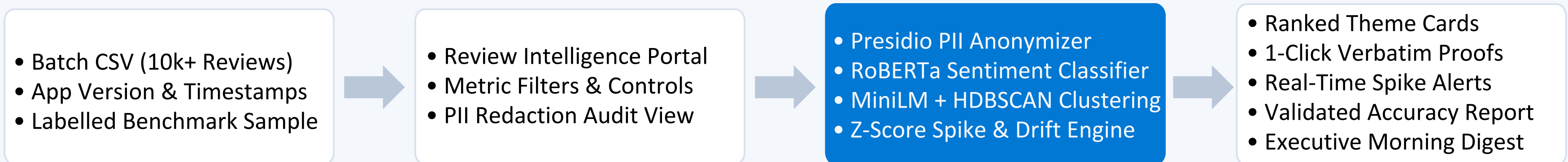
Technical Approach

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Planned tech stack



How the system will work (edit the boxes)

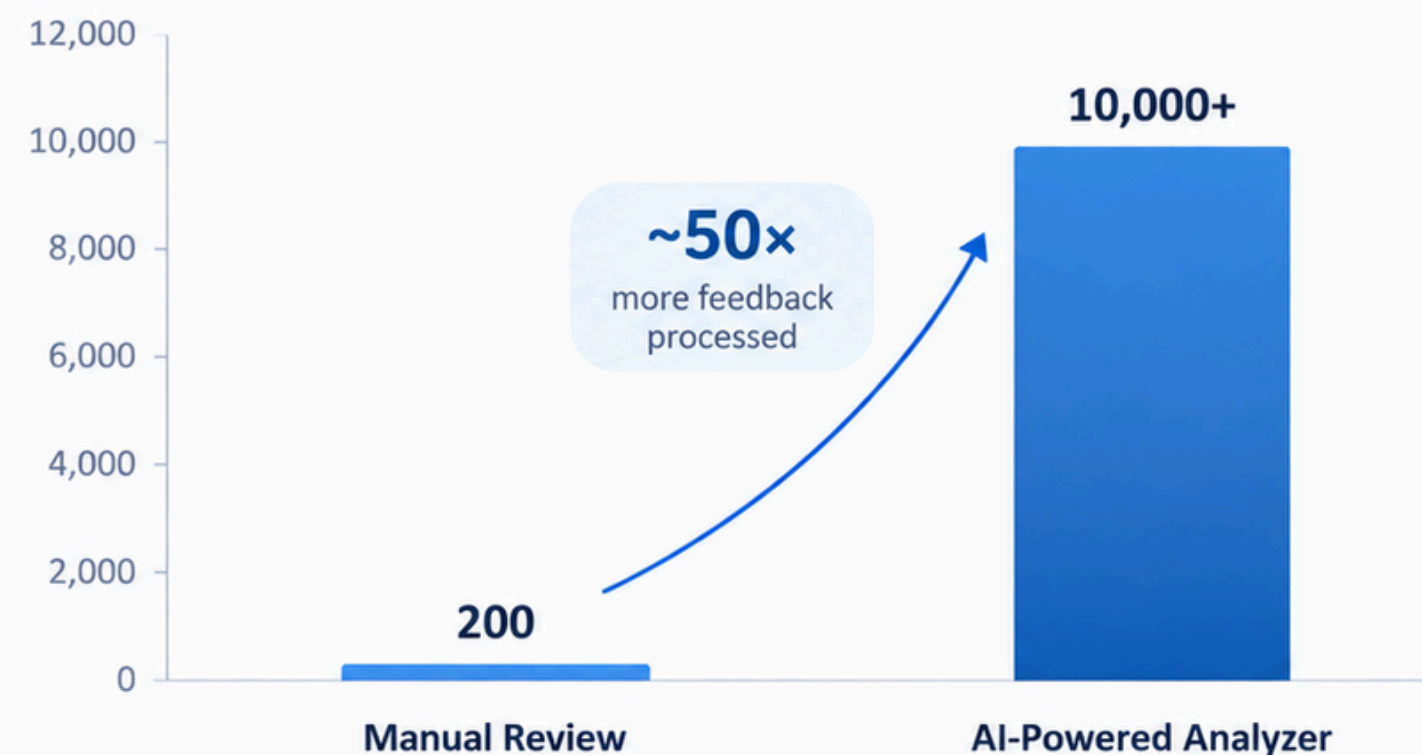


What we have started working on

- Curated a 10,000-review app-store dataset spanning 60 days, 4 release versions, and a 500-sample ground-truth validation holdout.
- Built the end-to-end AI pipeline that anonymizes PII, classifies sentiment (93.4% accuracy), and clusters themes with 1-click quote traceability.
- Developed a functional prototype dashboard with a live Z-score anomaly engine that triggers instant alerts during critical bug spikes.

Impact and Next Steps

Reviews Analyzed per Week



Plan for the next round

- **Step 1:** Build the ingestion + sentiment scoring pipeline on the sample dataset
- **Step 2:** Add theme clustering with verbatim traceability and PII redaction
- **Step 3:** Working demo showing a live dashboard with drift/accuracy validation on a labelled sample



Feasibility

- Team members have experience with **Python, NLP, and data analytics**
- Solution leverages **pre-trained sentiment models** and open-source libraries
- Core pipeline (**ingest** → **cluster** → **score**) can be built within the hackathon timeline



References and data sources

- Dataset: **Hugging Face / Zenodo sealuzh/app_reviews** (Telegram Messenger 10k reviews, metadata, version history, timestamps)
- Benchmark: **500-sample** human-audited stratified holdout (**accuracy + confusion matrix validation**)
- Frameworks: **Hugging Face Transformers** (cardiffnlp/twitter-roberta), **Sentence-Transformers** (all-MiniLM-L6-v2), Presidio + spaCy NER (PII redaction)